PORTFOLIO PROJECT · SQL + TABLEAU

What Drove AdventureWorks' 180% Revenue Growth?

A financial analysis of 60,398 online sales records and the company's 2011 budget, using SQL Server and Tableau

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Swipe for the findings →

5 Things the Data Shows

AdventureWorks, a bike retailer: full years 2011–2013

1

Revenue tripled in 2013

$5.8M → $16.4M, while gross margin held at ~41%

2

Growth came from volume, not price

More bikes sold; average price fell in every bike type

3

US & Australia = 63% of revenue

Australia matches the US with far fewer, bigger orders

4

Repeat customers = 77% of revenue

Only 37% of customers, but they spend 5–8x more

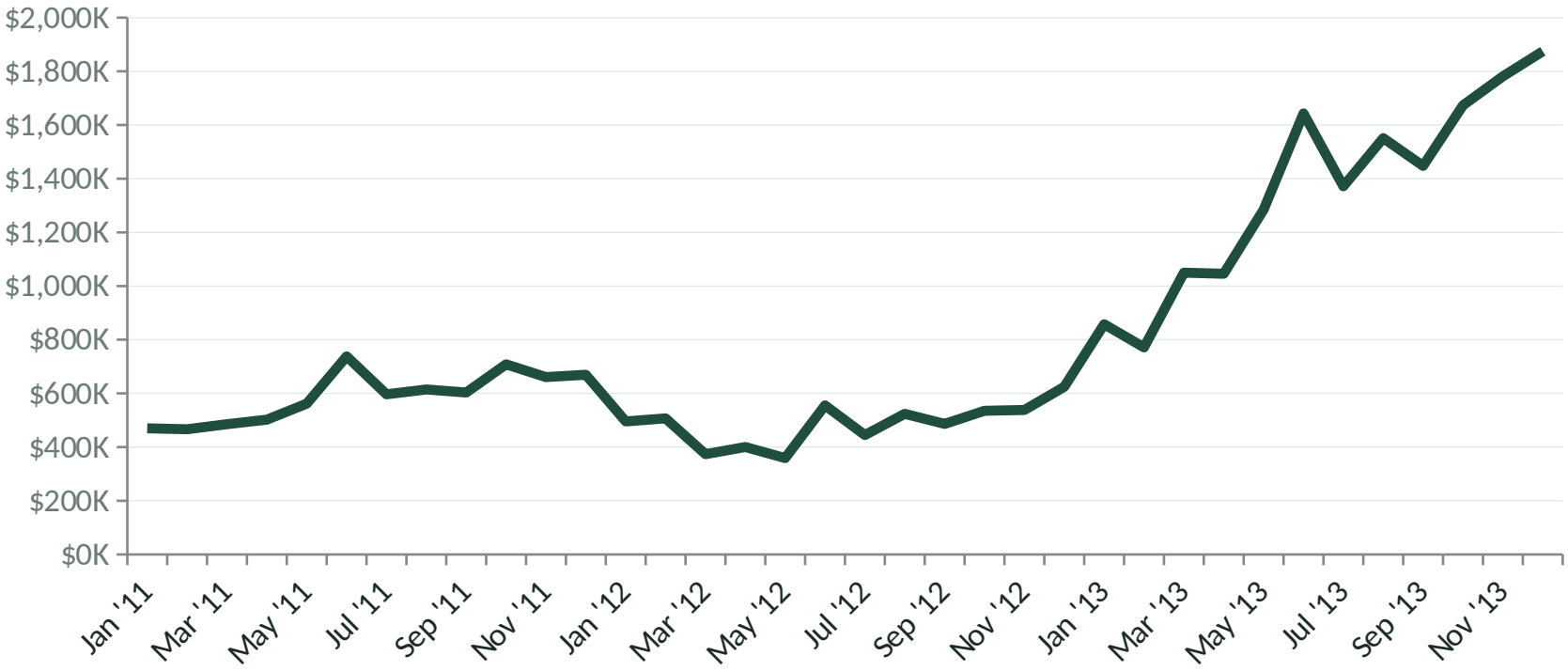
5

2011 profit missed budget by 35%

Revenue beat plan, but costs ran $4.0M (25%) over

Revenue Tripled in 2013

December 2013 brought in 3x December 2012, driven by 6.5x more orders



$470K

monthly revenue, Jan 2011

21,289

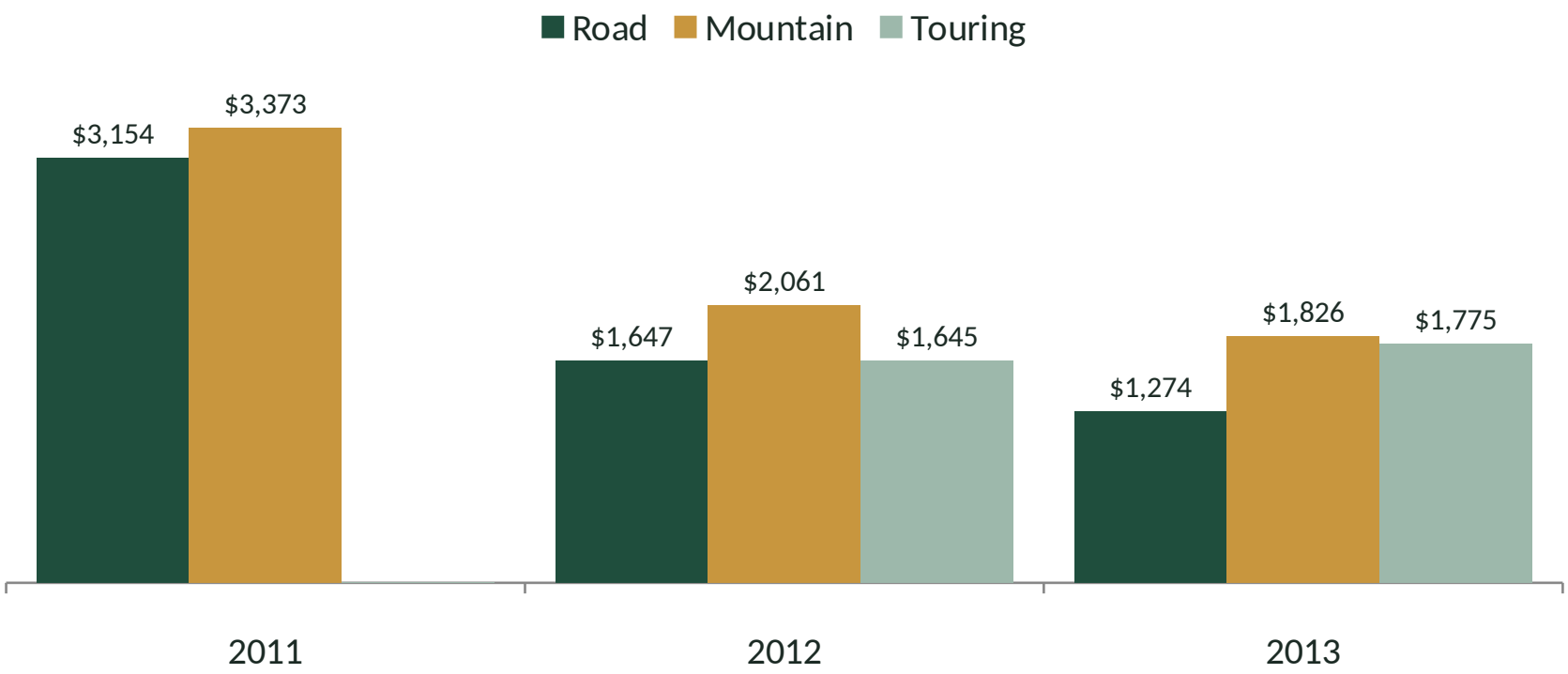
orders in 2013, up from 3,269 in 2012

$1.87M

monthly revenue, Dec 2013

Growth Came From Volume, Not Price

Average price fell in every bike type as cheaper models sold in far greater numbers



−60%

Road bike average price, 2011 → 2013

4.4x

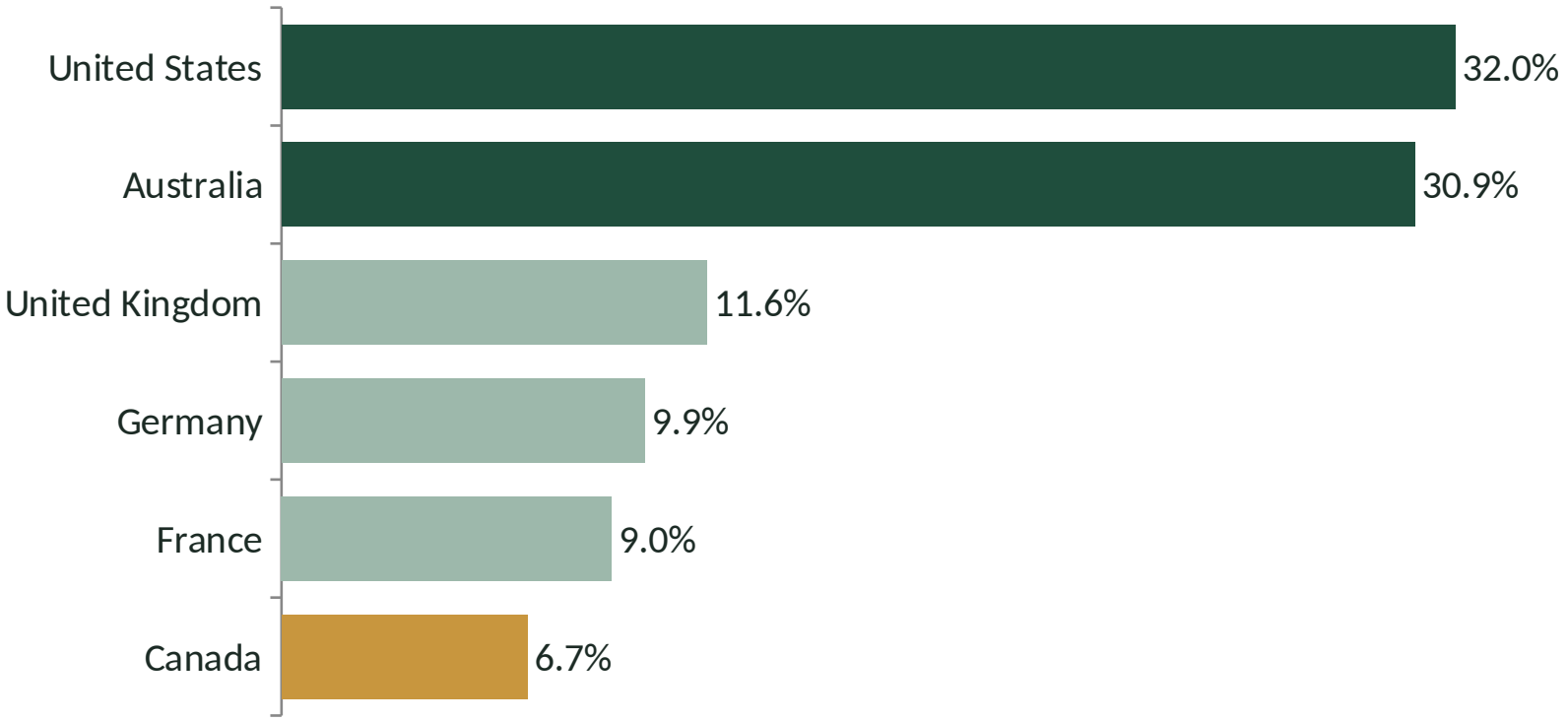
more bikes sold: 2,216 → 9,706

45%

Mountain bike margin, which kept overall margin at ~41%

US & Australia Earn 63% of Revenue

Margin is ~41% everywhere, so the gaps come from what customers buy, not discounts



$1,378

Australia's average order, the highest

$615

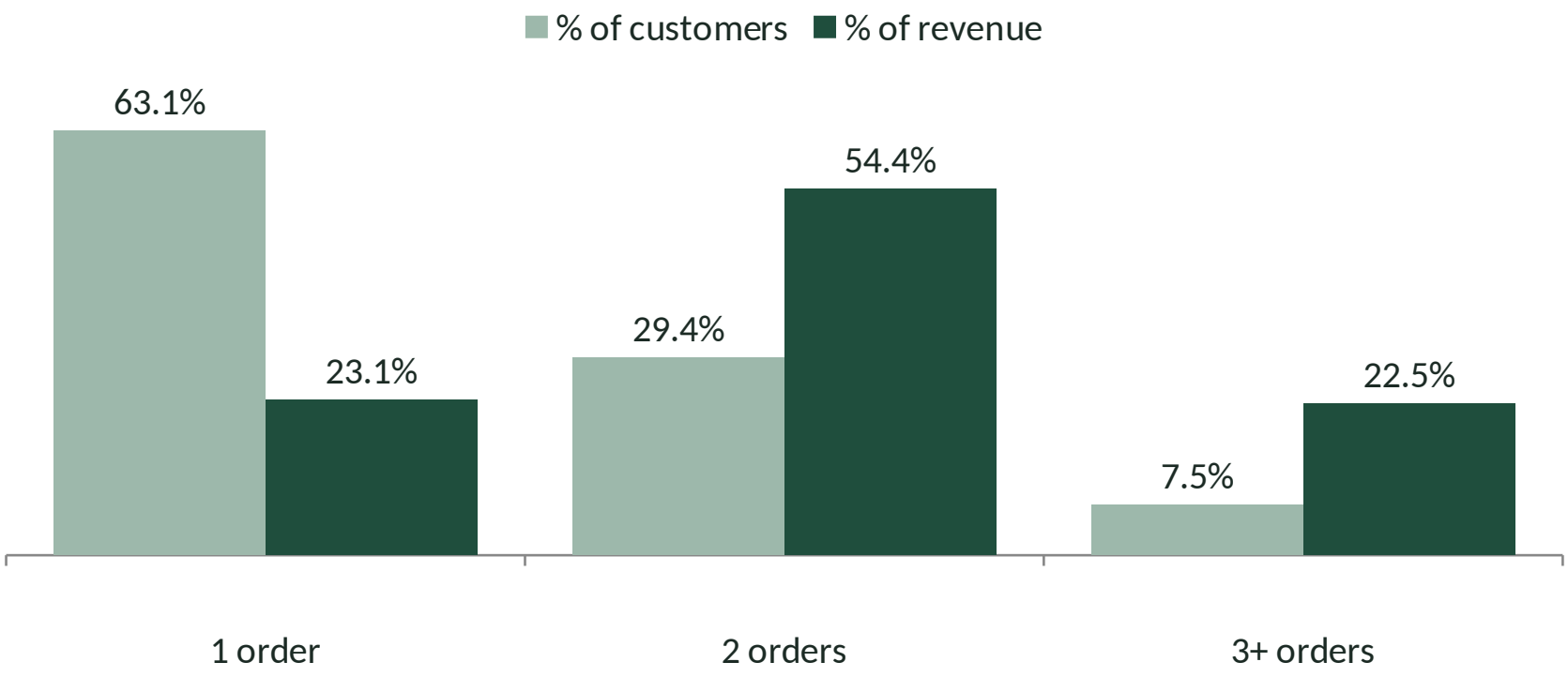
Canada's average order: 12% of orders, 7% of revenue

2 markets

drive nearly two-thirds of sales

Repeat Customers Bring In 77% of Revenue

Most customers buy once; the ones who come back spend 5–8x more



63%

of customers bought once: just 23% of revenue

$3,009

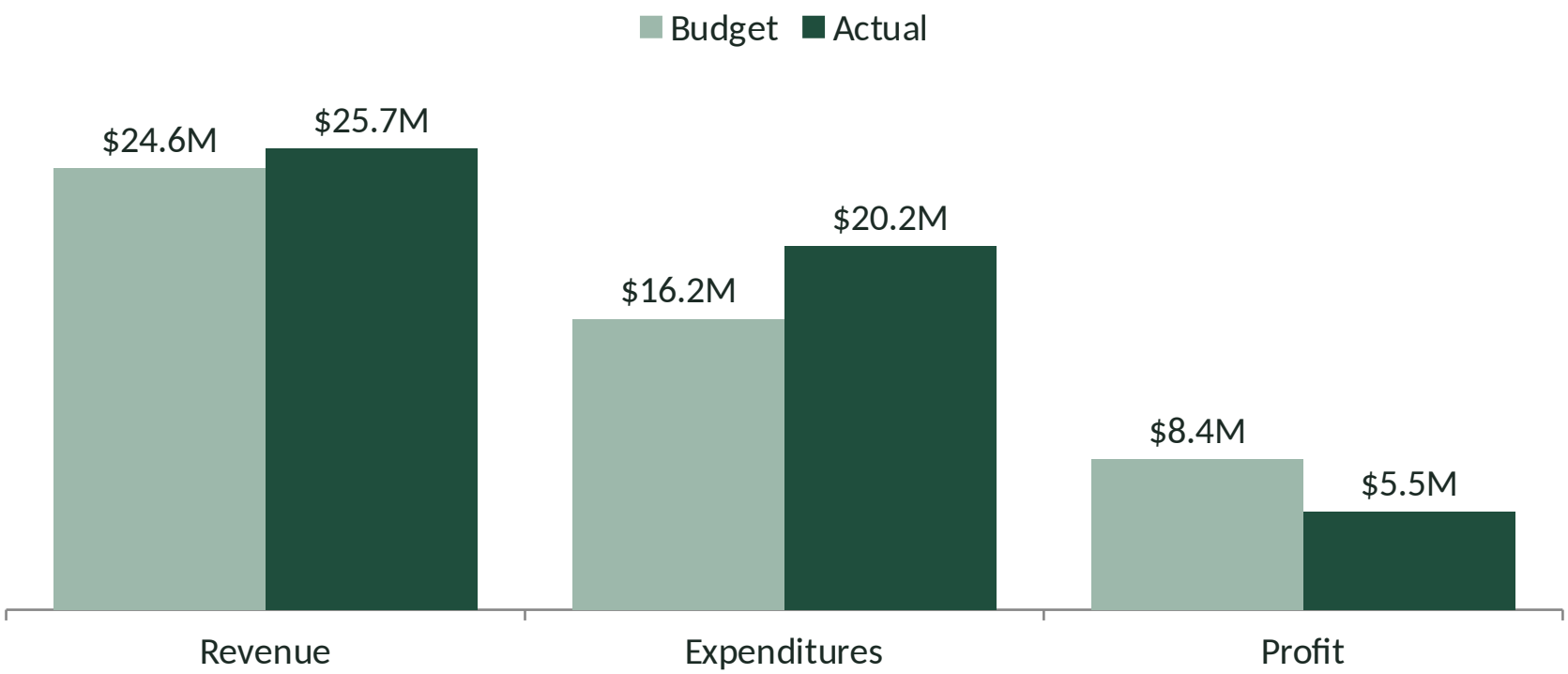
avg spend with 2 orders vs. $595 with 1

$4,888

avg spend of the most loyal 7.5%

2011: Revenue Beat Budget, Profit Missed by 35%

Costs ran $4.0M (25%) over plan; 89% of that came from two causes



$1.86M

in taxes that were never budgeted

+29%

people costs over plan ($1.71M)

–3.8%

cost of sales came in under budget

What I'd Recommend

Turning the findings into actions

Win the second purchase

Follow-up offers after a first order. Repeat buyers spend 5x more, so this is the biggest growth lever.

Push high-margin products

Promote Mountain bikes (45% margin) and accessories (63%) to protect margin as prices fall.

Grow order value in Canada

Lots of customers, small orders: bundle bikes with accessories to lift the $615 average.

Fix the budget process

Plan for taxes and set headcount budgets from realistic hiring assumptions.

How I Built It

From raw tables to a published dashboard

SQL Server	9 queries across 60,398 sales rows: multi-table joins, CTEs, window functions and conditional aggregation
Data checks	Found partial years (3 and 28 days) and excluded them; reconciled 29 expense variances to the $4.0M total
Tableau Public	Interactive dashboard: monthly trend, country map, customer segments, pricing and budget vs. actual
Analysis	Descriptive KPIs (revenue, gross margin, AOV, share) plus variance analysis, the core of FP&A work

SEE THE INTERACTIVE DASHBOARD

Growth came from volume. The next gains are in repeat customers, margin mix and better budgeting.

Tableau Public: public.tableau.com/app/profile/sana.husain5685

Link to the full dashboard and SQL is in the post.

Sana Husain · Open to analyst roles